

affirmoTM tutorial 1: getting started

What the model does, installing it, and a first run

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1 What affirmoTM does

affirmoTM compares two arms, **iABC** and **usual care**, in a cohort of patients with atrial fibrillation.

- **Yearly cycles:** each patient is followed one year (one *cycle*) at a time. In every year they are in one health state.
- **Transitions:** the next year's state is drawn from a transition matrix, one for each arm.
- **Costs and QALYs:** each year in a state carries a cost and a QALY weight.
- **Results:** adding these up gives each arm's costs and QALYs, the difference between the arms, the ICER and the cost-effectiveness (CE) plane.

A run of the model:

1. reads the input files from **inputs/** and checks them;
2. gives each patient a state for the year they enter;
3. moves every patient through each arm, year by year, until they die or the time horizon ends;
4. gives every year its cost and QALY, discounted;
5. adds up each arm and compares the arms.

The whole run is repeated several times with different random numbers; each run is one point on the CE plane.

1.1 Health states

The example files use these states. Your own files can use other states: the package reads them from the transition matrices.

State	Meaning
NoEvent	No stroke or major bleed
AIS, AHS, AB	Acute ischaemic stroke, haemorrhagic stroke or major bleed this year
PIS, PHS, PB	Past ischaemic stroke, haemorrhagic stroke or major bleed
AIS_AB, AHS_AB	Acute stroke and acute bleed this year
PB_AIS, PB_AHS	Past bleed and acute stroke this year (AIS_PB and AHS_PB in Figure 1)
PS_AB	Past stroke and acute bleed this year
PS_PB	Past stroke and past bleed
CurrentDeath	Death without a new event that year
AB_Death, AHS_Death, AIS_Death	Death in the year of that acute event
AHS_AB_Death, AIS_AB_Death	Death in the year of both acute events

Acute means the event happens during the current year; *past* means it happened in an earlier year.

Figure 1 shows the possible moves between the living states. Death states are not drawn; every living state can lead to death.

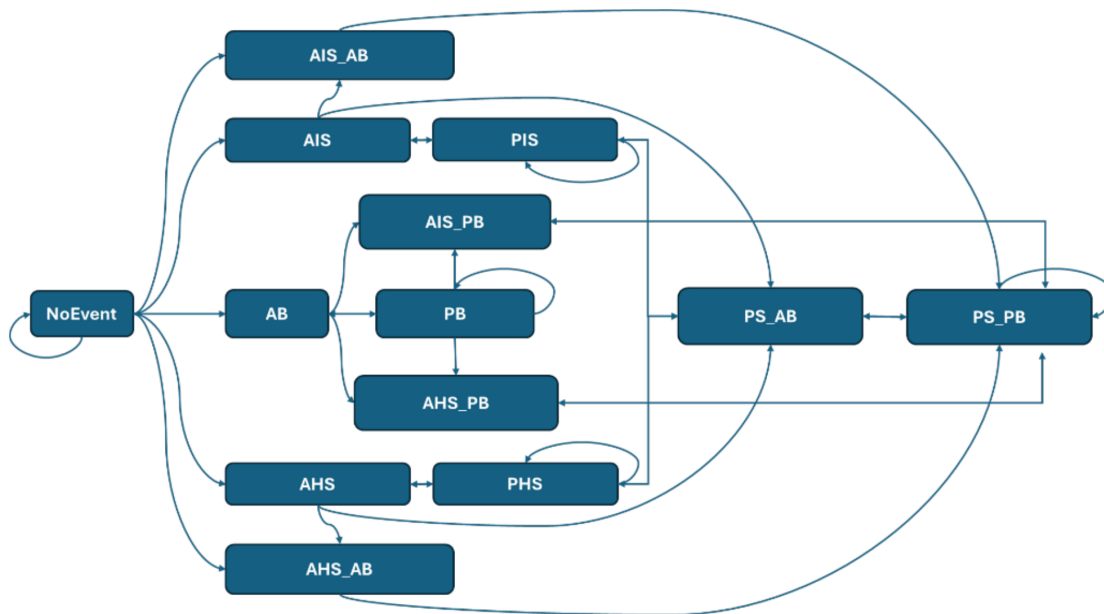


Figure 1: Disease states and allowed transitions for multimorbid AF patients (AFFIRMO).

2 Installing affirmoTM

You need R, and an editor such as RStudio or Positron. Install affirmoTM once; R also installs the packages it needs.

From GitHub:

```
install.packages("remotes")
remotes::install_github("RonyEA/affirmoTM")
```

From a copy of the repository on your computer:

```
remotes::install_local("path/to/affirmoTM")
```

Run the same line again to update.

3 An analysis folder

Each analysis lives in its own folder. Create an empty folder, for example `my_analysis`, and make it R's working directory: open it as a project in RStudio or Positron, or use `setwd()`. `affirmoTM` reads its inputs from `inputs/` and writes its results to `outputs/` in that folder.

Load the package and copy the example input files into the folder:

```
library(affirmoTM)
```

```
tm_example_inputs()
list.files("inputs")
```

```
[1] "cohort.csv"           "costs.csv"
[3] "entry_states.csv"     "intervention_cost.csv"
[5] "README.md"           "settings.yaml"
[7] "transition_matrix_iabc.csv" "transition_matrix_usual_care.csv"
[9] "utilities.csv"
```

The folder now looks like this:

```
my_analysis/
|-- inputs/           the files you supply (here, the examples)
|   |-- settings.yaml run settings
|   |-- transition_matrix_usual_care.csv
|   |-- transition_matrix_iabc.csv
|   |-- cohort.csv    the patients
|   |-- entry_states.csv the state in each patient's first year
|   |-- costs.csv     the cost of a year in each state
|   |-- utilities.csv  the QALY weight of a year in each state
|   |-- intervention_cost.csv the cost of delivering iABC
|   |-- README.md     describes the example files
`-- outputs/         written by every run
```

`tm_example_inputs()` never replaces files that already exist, unless you ask it to with `tm_example_inputs(overwrite = TRUE)`.

4 A first run

`tm_run()` reads every input, runs the model and writes the results to `outputs/`:

```
results <- tm_run()
results
```

`<tm_results>` mean per arm over 100 runs (discounted costs and QALYs)

	arm	patients	cost_total	qaly_total	intervention_cost_total
	<char>	<num>	<num>	<num>	<num>
1:	iabc	5000	65493591	18004.50	187494.6
2:	usual_care	5000	66035435	17980.58	0.0
	cost_per_patient	qaly_per_patient	intervention_cost_per_patient	ischaemic_stroke	

	<num>	<num>	<num>	<num>
1:	13098.72	3.601	37.499	429.23
2:	13207.09	3.596	0.000	461.93
	haemorrhagic_stroke	major_bleed	deaths	
	<num>	<num>	<num>	
1:	121.54	147.47	1850.85	
2:	129.50	160.45	1853.00	

iABC minus usual care (mean over runs, per patient):

Incremental cost: €-108.37 (intervention cost: €37.5)

Incremental QALYs: 0.00478

ICER: iABC dominant (cheaper, more QALYs)

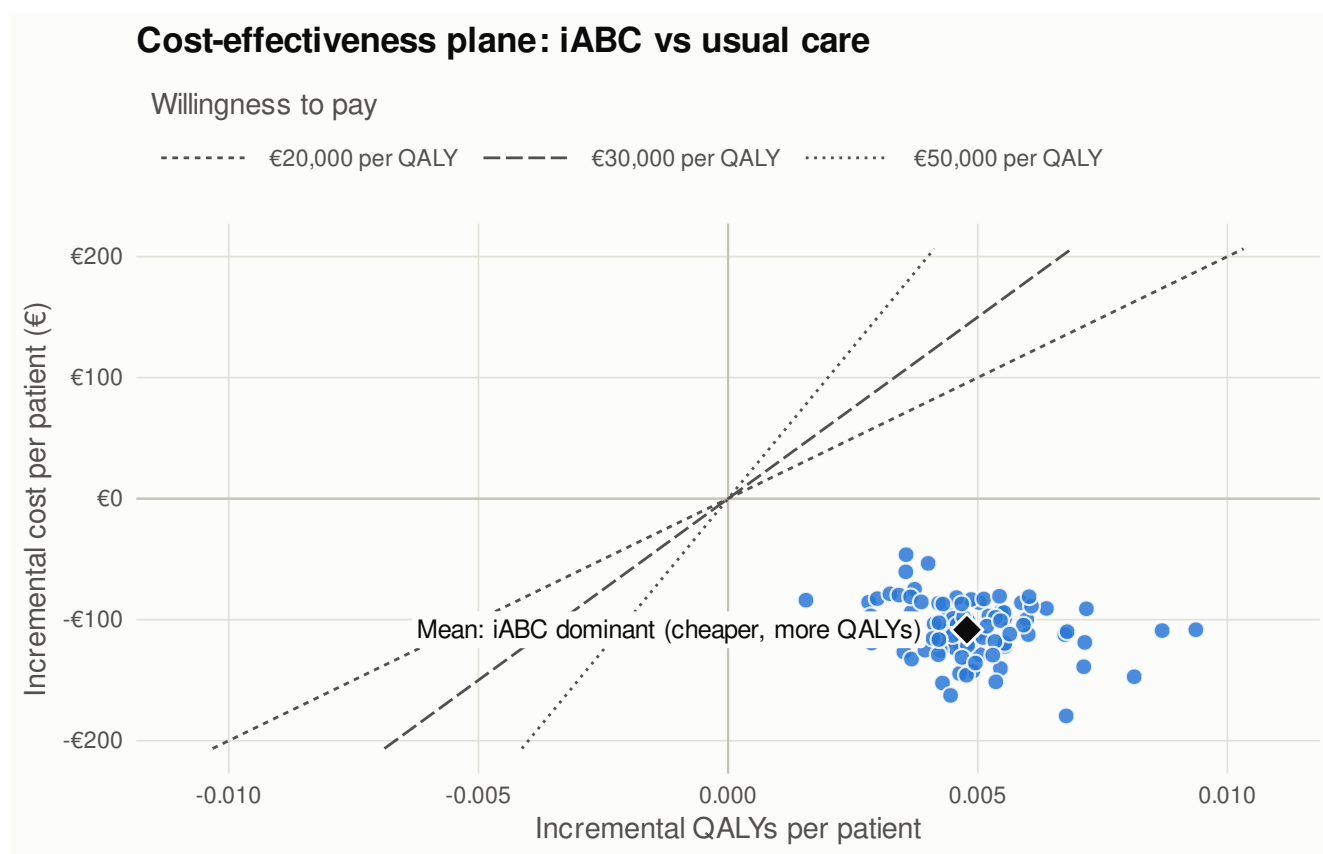
Difference in events (whole cohort): ischaemic_stroke -32.7, haemorrhagic_stroke -8.0, major_bleed
 ↪ -13.0, deaths -2.1

Written to outputs/: summary.csv, summary_by_run.csv, comparison.csv, comparison_by_run.csv, ce_plane.png

The printout shows each arm's mean costs, QALYs and events over the runs, then iABC minus usual care: the incremental cost, the incremental QALYs and the ICER.

The cost-effectiveness plane:

```
plot(results)
```



5 Where next

Tutorial	What it covers
2. Input files	Each input file: its layout, its rules, and how to look at it
3. Settings	Each setting in <code>settings.yaml</code> , and how to change them
4. Running the model	What <code>tm_run()</code> does, and the steps of a run one at a time
5. Results	The results object, the comparison, the CE plane and the output files
6. Walkthrough	One run followed in detail: patients, costs, QALYs and how the arms behave over time

Each function also has a help page, for example `?tm_run`.